Theorem 1. *In the quantum state discrimination problem, given the same set of mixed quantum states* $\{\rho_1, \rho_2, \ldots, \rho_M\}$ *and the same prior probabilities* $\{p_1, p_2, \ldots, p_M\}$, *the probability of correctly identifying the given state in the optimal MED scenario is identical to that in the optimal MED+ scenario; i.e.,* $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED+}}$. *Furthermore,* $\Pi_{M+1} = 0$ *when the MED+ scenario becomes optimal.*

Proof.

Let POVM_M be the collection of POVMs that each POVM has M POVM elements, and here we allow a POVM element to be zero. Let $(P_{\text{succ}})^P$ denote the probability of successful discrimination of some POVM P or the POVM derived from some method P. We know that $(P_{\text{succ}})^{\text{MED}} >= (P_{\text{succ}})^P, \ \forall\ P \in \text{POVM}_\text{M}$.

1. $(P_{\text{succ}})^{\text{MED}} \leq (P_{\text{succ}})^{\text{MED+}}$
   Assume POVM_{MED} is an optimal solution for MED, and $\text{POVM}_{\text{MED}} \equiv \{\Pi_1, \Pi_2, \ldots, \Pi_M\} \in \text{POVM}_\text{M}$. We can construct $P \equiv \left\{\Pi'_1, \Pi'_2, \ldots, \Pi'_{M+1}\right\} \in \text{POVM}_{\text{M+1}}$ by

$$\Pi'_1 = \Pi_1,\ \Pi'_2 = \Pi_2,\ \ldots,\ \Pi'_M = \Pi_M,\ \Pi'_{M+1} = 0$$

   Then,

$$\begin{aligned}(P_{\text{succ}})^{\text{MED+}} \geq (P_{\text{succ}})^P &= \sum_{i=1}^{M} p_i \operatorname{tr}(\rho_i \Pi'_i) && (1)\\ &= \sum_{i=1}^{M} p_i \operatorname{tr}(\rho_i \Pi_i) && (2)\\ &= (P_{\text{succ}})^{\text{MED}} && (3)\end{aligned}$$

2. $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{\text{MED+}}$
   Assume $\text{POVM}_{\text{MED+}}$ is an optimal solution for MED+, and $\text{POVM}_{\text{MED+}} \equiv \{\Pi_1, \Pi_2, \ldots, \Pi_{M+1}\} \in \text{POVM}_{\text{M+1}}$. We can construct $P \equiv \{\Pi'_1, \Pi'_2, \ldots, \Pi'_M\} \in \text{POVM}_\text{M}$ by

$$\Pi'_1 = \Pi_1,\ \Pi'_2 = \Pi_2,\ \ldots,\ \Pi'_M = \Pi_M + \Pi_{M+1}$$

Then,

$$
\begin{aligned}
(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{P} &= \sum_{i=1}^{M} p_i \operatorname{tr}(\rho_i \Pi_i') && (4)\\
&= \sum_{i=1}^{M} p_i \operatorname{tr}(\rho_i \Pi_i) + p_M \operatorname{tr}(\rho_i \Pi_{M+1}) && (5)\\
&= (P_{\text{succ}})^{\text{MED+}} + p_M \operatorname{tr}(\rho_i \Pi_{M+1}) && (6)\\
&\geq (P_{\text{succ}})^{\text{MED+}} && (7)
\end{aligned}
$$

3. By 1. and 2., we can see that $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED+}}$.

4. To make the equal sign in Eq (7) hold, $\Pi_{M+1} = 0$.

□

From Theorem 1, we can easily prove the following corollary.

Corollary 1. *In the quantum state discrimination problem, given the same set of mixed quantum states $\{\rho_1, \rho_2, \ldots, \rho_M\}$ and the same prior probabilities $\{p_1, p_2, \ldots, p_M\}$, the probability of correctly identifying the given state in the optimal MED scenario is always equal to or larger than that in the Optimal UQSD scenario; i.e., $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{\text{OptUQSD}}$.*

Proof. If we see the MED+ problem and the Optimal UQSD problem as an convex optimization problem, then they have the same variables and the same objective function (P_{succ}), with the Optimal UQSD problem posing extra constraints on the error terms ($P_{\text{err}} = 0$). This means that the optimal solution of OptUQSD is always a feasible point of MED+, so $(P_{\text{succ}})^{\text{MED+}} \geq (P_{\text{succ}})^{\text{OptUQSD}}$. Then $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED+}} \geq (P_{\text{succ}})^{\text{OptUQSD}}$. □